

ADAM TROTT

[private phone] | [university email]

EDUCATION

University of Massachusetts Dartmouth

Ph.D. in Engineering and Applied Science (EAS), Computer Science and Information Science (CSIS) Curriculum Option

Expected May 2030 | Graduate GPA: 4.0/4.0

[private government application credential - withheld] | Jan. 2026-Present

Selected Graduate Coursework: Network Security & Data Assurance; Secure Software Development; Fundamentals of Deep Learning; Mathematics of Deep Learning; Scientific Machine Learning (in progress, Fall 2026)

University of Massachusetts Dartmouth

B.S. in Computer Science, Cybersecurity concentration | May 2025

GPA: 3.379/4.0 | Cum Laude

RESEARCH EXPERIENCE

Research Assistant - University of Massachusetts Dartmouth | Dartmouth, MA

Jun-Aug 2025; Jun-Aug 2026

- Developed and tested an operational-AI red-teaming framework on an autonomous vehicle with cybersecurity, computer science, and electrical-engineering researchers using ROS2 and Gazebo.

DQNGuard: Towards Open-World RF Preliminary-Action Detection

First author | IEEE MILCOM 2026, accepted/to appear

- Designed and implemented the end-to-end DQNGuard pipeline for open-world RF preliminary-action detection, including a multi-domain CNN and open-set decision layer; benchmarked against VarMax and a DQN-IDS-style baseline.
- Built an over-the-air SDR capture, preprocessing, and dataset pipeline for WiFi, Bluetooth, and Zigbee RF signals; designed, trained, and evaluated experiments across five preliminary actions: Scan, Burst, Sustain, Hop, and Replay.
- Achieved 0.865 mean unknown-class F1 under a 5% known-rejection calibration budget, compared with 0.745 for VarMax and 0.701 for the DQN-IDS-style comparison head; designed and analyzed a 20-condition Target-Surrogate Matrix.

Model Evaluation for Radio-Frequency Signal Modulation Classifiers in the Existence of Novel Samples

First author | HICSS-59, published 2026; presented Jan. 8, 2026

- Developed and evaluated a class-conditioned RF open-set-recognition pipeline using an eight-channel CNN over raw I/Q, FFT, DCT, and polar representations.
- Evaluated nine open-set folds on RadioML 2018.01A, withholding one selected modulation as unknown per fold and running 30 runs per fold.

SELECTED PUBLICATIONS

Trott, A., Popillo, C., Bastian, N. D., Zhou, R., and Kul, G. "DQNGuard: Towards Open-World RF Preliminary-Action Detection." IEEE MILCOM 2026, accepted/to appear.

Trott, A., Thompson, H., and Kul, G. "Model Evaluation for Radio-Frequency Signal Modulation Classifiers in the Existence of Novel Samples." Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS-59), 2026.

TECHNICAL SKILLS

AI / Machine Learning: Python, PyTorch, Hugging Face, Jupyter; CNNs, transformers, open-set recognition, novelty/OOD detection, model calibration, experimental design, quantitative/statistical evaluation

RF / Signal Processing: software-defined radio (SDR), RF signal analysis, complex I/Q, FFT, DCT, and polar representations; over-the-air WiFi, Bluetooth, and Zigbee data workflows

Cybersecurity / Systems / Software: Linux/WSL, Docker/Docker Compose, networking, OpenSSL, Autopsy, Burp Suite, Flask, MongoDB, Nginx, ROS2, Gazebo

SELECTED TECHNICAL AND TEACHING EXPERIENCE

Lead Backend Developer & Project Manager - UMass Dartmouth Senior Design Capstone

Client: NUWC Newport | Sep. 2024-May 2025

- Led backend development and project management for a university team building a file-storage, viewing, and search system for a NUWC Newport client; coordinated technical work and delivery with the client.

Teaching Assistant - Cyber Defense and Operations - University of Massachusetts Dartmouth | Jan.-May 2026

- Designed and graded practical cybersecurity assignments involving risk analysis, cryptography, WebGoat SQL injection, and Burp Suite access-control/authentication attacks.
- Authored a Caesar Cipher/DDES/3DES cryptanalysis project requiring candidate-key generation, brute-force search, plaintext verification, performance measurement, analytical reporting, and a recorded demonstration.

Teaching Assistant - Digital Forensics - University of Massachusetts Dartmouth | Sep.-Dec. 2025

- Created Autopsy disk-image and Docker-based macOS log-analysis labs and guided students through query construction, visualization, and incident-style reporting.